

Solutions in Motion™

Kinematic Restoration

The art and science of restoring knee function



Johnson & Johnson
MedTech

What is Kinematic Restoration?

Kinematic Restoration is the art and science of restoring knee function.



Technique: From pre-op through post-op, it provides a holistic approach that works in harmony with kinematic and inverse kinematic alignment philosophies

Implant: The kinematically-advanced ATTUNE™ Knee has been verified* to work with kinematic alignment techniques within tested boundaries¹⁻⁴

Technology: The VELYS™ Robotic-Assisted Solution equips surgeons with real-time insights to optimize implant placement, preserve soft tissue envelope, predict joint stability and work toward returning knee function⁵

*Based on in vitro studies

This powerful combination can reduce soft tissue releases⁵ to restore knee function and may accelerate patient recovery.⁵



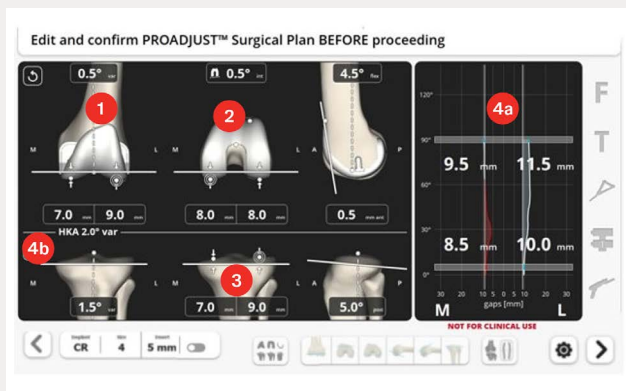
Technique

There are two defined kinematic philosophies within Kinematic Restoration:
Kinematic Alignment (KA) and Inverse Kinematic Alignment (iKA)

KA

Kinematic Alignment is a femur-first technique that aims to position the femoral and tibial components to resurface the knee and restore the natural joint line accounting for cartilage loss that occurs during disease progression.⁶

KA systematic planning process⁷



- 1 Distal femoral resection depth**
 - Recreate lateral distal femoral angle (LDFA)
- 2 Posterior condyle resection depth**
 - Resurface posterior femoral condyles
- 3 Proximal tibia resection depth**
 - Recreate medial proximal tibia angle (MPTA)
- 4a ACCUBALANCE™ Graph**
 - Utilize soft-tissue data if boundaries are used
- 4b HKA (long leg alignment)**
 - Utilize HKA target if boundaries are used

iKA

Inverse Kinematic Alignment is a tibia-first, balanced technique where the aim is to resurface the tibia, maintaining the joint line obliquity and then set the femoral position to balance the knee in flexion and extension.^{5,8}

iKA systematic planning process⁹



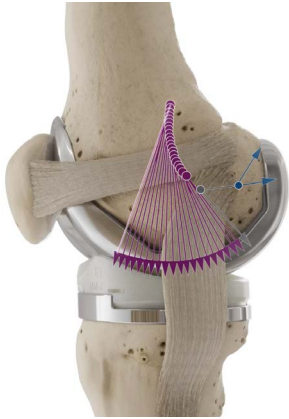
- 1 Proximal tibia resection depth**
 - Recreate medial proximal tibial angle (MPTA)
- 2 Soft-tissue ACCUBALANCE™ Graph**
 - Initial priority when balancing femur post tibia resection
- 3 HKA (long leg alignment)**
 - HKA target/boundary
- 4 Distal femoral resection depth**
 - Target recreating lateral distal femoral angle (LDFA) or matching resection depth to component thickness
- 5 Medial posterior femoral condyle resection depth**
 - Target resection depth to match resection depth to component thickness

Kinematic Restoration aims to minimize soft tissue releases and surgical complications, to facilitate faster recovery.^{5,10}

Implant

ATTUNE™ Knee - A kinematically advanced implant

STABILITY IN MOTION™



ATTUNE GRADIUS™ Curve
+
Medial Stabilized Insert

ATTUNE Knee Femoral Curve¹¹⁻¹⁵

- Gradually reducing femoral radius for smooth transition throughout the gait cycle
- Maintains tibio-femoral conformity from extension to flexion
- Appropriately tension the MCL throughout full range of motion (ROM)

Why it matters: Eliminates sudden paradoxical anterior translation, replicating motion observed in native knees.

MCL = Medial Collateral Ligament

ATTUNE Medial Stabilized Knee System^{12,14,16-18}

- Reduces A/P motion without abrupt direction changes
- Increases conformity on the medial compartment
- Lateral compartment is flatter to encourage natural femoral rollback

Why it matters: Medial knee stability during TKA has been shown to improve patient satisfaction and knee function.

Advanced patellofemoral articulation



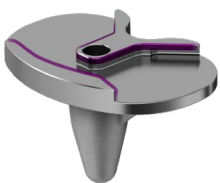
GLIDERIGHT™ Articulation

ATTUNE Knee Patellofemoral Articulation¹⁹⁻²⁶

- Addresses alterations in trochlear orientation seen in kinematic alignment techniques
- Trochlear angle varies by size to accommodate varying patient Q-angles
- Funnel shape from 0-45 degrees allows the patient's soft tissues to drive the position of the patella in extension and provides a smooth transition into the center of the groove

Why it matters: Accommodates patient variation and strikes a balance between accommodating the natural soft tissues while providing the appropriate amount of stability in the implant.

Optimal tibiofemoral articulation



LOGICLOCK™ Tibial Base

Fixed Bearing Tibial Base²⁷⁻³²

- Central locking tibial tray and insert design allows the femoral component and tibial insert to match 1:1 every time
- Accommodates 2-up and 2-down sizing between the tibial insert and tibial base
- Available on all FB tibial designs & compatible with CR, MS, PS inserts

Why it matters: Provides option to independently size the femur and tibia while maintaining optimized articulation, stability, and rotational freedom throughout the range of motion for every patient.

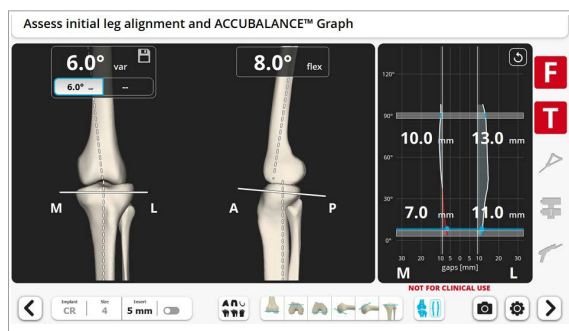
FB = Fixed Bearing • CR = Cruciate Retaining • MS = Medial Stabilized • PS = Posterior Stabilized

Technology

VELYS™ Robotic-Assisted Solution complements kinematic philosophies to help enable precise and accurate ATTUNE™ Knee placement for each patient^{7, 33-34*}

*Based on study comparing accuracy of VELYS System vs. Manual

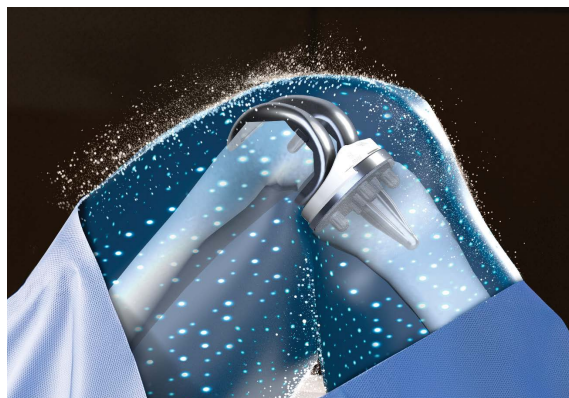
Real-time personalized insights



Informed Decision Making^{7,35-37}

- Freedom to place components to optimize alignment philosophies
 - KA surgeons resurface the femur and iKA surgeons resurface the tibia to maintain joint line obliquity
 - Long leg alignment and correctability influence surgeon decision making
- Surgeons may utilize resection locks to avoid compromising on long leg alignment, soft tissue balance, and resection depth goals

Why it matters: Allows for informed decision-making that accounts for soft-tissue tension to achieve Joint Line Obliquity (JLO), balance targets, and correctability goals.



Accurate Implant Placement^{7,9,33,37-39}

- ACCUBALANCE™ Graph Soft-tissue and Gap Data
 - Surgeons have the ability to use soft tissue and gap data throughout the entire range of motion to better inform decision making
 - iKA surgeons use this soft tissue data to reproducibly position their femoral components to preserve the native soft tissue envelope.

Why it matters: Robotic-assisted TKA lowered component deviations and outliers in comparison to manual TKA, resulting in more accurate implant alignment without compromising the soft tissue structures and balance of the knee.

Precise saw navigation



Intraoperative Landmark Registration^{7,33,34**}

Surgeon-dictated coronal and sagittal plane resection depths and alignment to accommodate soft and hard structure.

Why it matters: The VELYS Knee system helps to execute precise and reproducible surgeon-controlled cuts that automatically account for the thickness of the ATTUNE Knee implant selected.

** Pre-clinical test data/results may not necessarily be indicative of human clinical performance (or outcomes).

References

- Johnson & Johnson and its affiliates. Vivo Knee Simulator wear test—anatomical alignment ATTUNE TKA, 2017. Windchill #103384220.
- Johnson & Johnson and its affiliates. ATTUNE Fixed Bearing: intraoperative ROM simulation and pull-off test with lipid and marrow, 2017. Windchill #103397037.
- Johnson & Johnson and its affiliates. ATTUNE porous tibia micromotion with anatomic and mechanical alignment, 2021. Windchill #103839160.
- Johnson & Johnson and its affiliates. Anatomic alignment surgical technique and patellar tracking validation report, 2017. Windchill #103335944.
- Spitzer A, Gorab R, Barrett W, et al. Robotic-assisted total knee arthroplasty reduces soft-tissue releases which improves functional outcomes: a retrospective study. *Knee*. 2024;49:52-61.
- Howell S, Roth J, Hull ML. Kinematic Alignment in Total Knee Arthroplasty Definition, History, Principle, Surgical Technique, and Results of an Alignment Option for TKA. *Arthropeadia*. April. 2014:44-53.
- Johnson & Johnson and its affiliates. VELYS™ Robotic-Assisted Solution for Total Knee. Surgical Technique, Femur First Approach (November 2024). Windchill Report 103744585 Rev J.
- Vigdorchik JM, Wakelin EA, Koenig JA, Ponder CE, Plaskos C, DeClaire JH, et al. Impact of Component Alignment and Soft Tissue Release on 2-Year Outcomes in Total Knee Arthroplasty. *J Arthroplasty* 2022.
- Johnson & Johnson and its affiliates. VELYS™ Robotic-Assisted Solution for Total Knee. Surgical Technique, Tibia First Approach (June 2024). Windchill Report 103744586 Rev. G.
- Huang P, Cross M, Gupta A et al. Early Clinical and Economic Outcomes for the VELYS Robotic-Assisted Solution Compared with Manual Instrumentation for Total Knee Arthroplasty. *J Knee Surg* 2024;37:864–872.
- Johnson & Johnson and its affiliates. Design verification study: forward dynamics analysis, DKB simulation of the ATTUNE MS and CR designs. 2021. Windchill. 103664945.
- List E, Schutz R, Angst M, Ellenberger L, Datwyler K, Ferguson SJ. Videofluoroscopic Evaluation of the Influence of a Gradually Reducing Femoral Radius on Joint Kinematics During Daily Activities in Total Knee Arthroplasty. *Journal of Arthroplasty*, 2020; 35 (10), 3010-3030.
- Pfzner T, Moewis P, Stein P, et al. Modifications of femoral component design in multi-radius total knee arthroplasty lead to higher lateral posterior femoro-tibial translation. *Knee Surg Sports Traumatol Arthrosc*. 2018;26(6):1645-1655. doi:10.1007/s00167-017-4622-7.
- Khasian M, Sharma A, Fehring TK, Griffin WL, Mason JB, Komistek RD. Kinematic Performance of Gradually Variable Radius Posterior-Stabilized Primary TKA During Various Activities: An In Vivo Study Using Fluoroscopy. *J Arthroplasty*. 2020;35(4):1101-1108. doi:10.1016/j.arth.2019.10.053.
- Takagi H, Asai S, Sato A, Maekawa M, Kawashima H, Kanzaki K. Case series report of navigation-based in vivo knee kinematics in total knee arthroplasty with a gradually reducing femoral radius design. *Ann Med Surg (Lond)*. 2017;17:33-37. Published 2017 Mar 29. doi:10.1016/j.jamsu.2017.03.032.
- Johnson & Johnson and its affiliates. Forward dynamics modeling analysis: multi-patient Design Verification Report. 2021. Project #103664920. Windchill 103845908.
- Tsukiyama H, Kuriyama S, Kobayashi M, et al. Medial rather than lateral knee instability correlates with inferior patient satisfaction and knee function after total knee arthroplasty. *Knee*. 2017;24(6):1478-1484. doi:10.1016/j.knee.2017.09.004.
- Mane AM, Komistek RD, Heldreth MA. The soft tissue harmony with the knee motion in Cruciate Retaining Arthroplasty, ORS-2021.
- Clary, C.W., Wright, A.P., Komosa, M.C., Maletsky, L.P. (2012b). The effect of patella medialization on patellofemoral kinematics after total knee replacement. 18th Congress of the European Society of Biomechanics (ESB); 29: 1262.
- Fitzpatrick CK, Clary CW, Rullkoetter PJ. The role of patient, surgical, and implant design variation in total knee replacement performance. *Journal of Biomechanics*. 2012;45:2092-2102.
- Mane, A, Boldt R, et al. Does Femoral Flange Design Influence PF Mechanics in PSA. ISTA. 2023. Poster #8257.
- Barrett D, Wright, A., ATTUNE Knee System: GLIDERIGHT Articulation Whitepaper. 2014. 0612-68-510.
- Deacy J, Fitzpatrick C. Comparison of Natural and Unresurfaced Patellofemoral Mechanics. ORS 2012. Poster #1062.
- Clary CW, Fitzpatrick CK, Maletsky LP, Rullkoetter PJ, Probabilistic Modeling of Fem Component Overhang. ORS Poster #1988. 2012.
- Fitzpatrick C, Baldwin M. Identifying Alignment Parameters Affecting Implanted Patellofemoral Mechanics. *Journal of Orthopaedic Research*. JULY 2012. 1167-75.
- Johnson & Johnson and its affiliates. IFU – ATTUNE Total Knee System - Fixed Bearing and Rotating Platform Bearing Knee. 0902-60-062,elFU Rev. A.
- Clary, C.W., Fitzpatrick, C.K., Maletsky, L.P., Rullkoetter, P.J. Improving dynamic mid-stance stability: an experimental and finite element study. ORS Annual Meeting, February 2012a Poster #1044.
- Clary CW, Fitzpatrick CK, Maletsky LP, Rullkoetter PJ. The influence of total knee arthroplasty geometry on mid-flexion stability: an experimental and finite element study. *J Biomech*. 2013;46(7):1351-1357. doi:10.1016/j.jbiomech.2013.01.025.
- Fitzpatrick, C.K., Clary, C.W., Rullkoetter, P.J. Post-cam engagement during dynamic activity with current posterior-stabilized TKR. 18th Congress of the European Society of Biomechanics (ESB), 2012a, 1700: 29.
- Heldreth M. ATTUNE LOGICLOCK Tibial Base Central Locking Design. White Paper. 2013 0612-67-513.
- Mane A, Rock M et al. Effect of Varying Femur-Insert Congruency, due to Sizing Philosophies, on Knee Mechanics. ORS Annual Meeting 2023. Paper No. 326.
- Mane A, Rock M, et al. Effect of Femur-Insert Size Mismatch on Tibiofemoral Joint Motions and Soft Tissue Structures. ISTA 2023. Poster #7505.
- Doan, G.W., et al., Image-Free Robotic-Assisted Total Knee Arthroplasty Improves Implant Alignment Accuracy: A Cadaveric Study. *J Arthroplasty*, 2022. 37(4): p. 795-801.
- Alton TB, Severson EP, Ford M, Leslie IJ, Lesko J, Diaz R, et al. Assessment of Accuracy and Early Outcomes During the Adoption Phase of a Novel Image-Free Robotic-Assisted System for Total Knee Arthroplasty. *British Orthopaedic Association*. Liverpool; 2023.
- Howell SM, Hull ML. Kinematic alignment in TKA: definition, surgical technique, and challenging cases. *Orthopedic Knowledge Online* 2012;10(7). https://www.researchgate.net/publication/235769696_Kinematic_Alignment_in_TKA_Definition_Surgical_Technique_and_Challenging_Cases.
- Winnock de Grave P, Kellens J, Luyckx T, Tampere T, Lacaze F, Claeys K. Inverse Kinematic Alignment for Total Knee Arthroplasty. *Orthopaedics & traumatology, surgery & research : OTSR* 2022;108(5):103305. <https://doi.org/10.1016/j.otsr.2022.103305>.
- Shichman I, Hadad A, Brandstetter AS, Ashkenazi I, Warschwaski Y, Gold A, et al. Better restoration of joint line obliquity in tibia first restricted kinematic alignment versus mechanical alignment TKA. *Archives of orthopaedic and trauma surgery* 2024. <https://doi.org/10.1007/s00402-024-05551-8>.
- Clatworthy M. Patient-Specific TKA with the VELYS Robotic-Assisted Solution. *Surg Technol Int* 2022;40:315-20. <https://doi.org/doi: 10.52198/22.ST140.OS156>.
- Courtis RP, Doan GW, Wyss JG, Leslie IJ, Green EW, Clary C. Assessment of Soft Tissue Following use of a Novel Concept for Robotic-Assisted Total Knee Arthroplasty. *Orthopaedic Research Society*. Virtual; 2021.

Important Information: Prior to use, refer to the instructions for use supplied with the device(s) for indications, contraindications, side effects, warnings and precautions.